

SIMATS ENGINEERING ASSESSMENT TOOL1

COURSE CODE: MLA 0202 **COURSE NAME:** Fundamentals of Machine Learning

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COURSE OUTCOME COVERED

***CO2:** Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 10 Total Marks:50 Annexure A

Analytical Problem Solving

Code of Conduct:

*I **Ch.B.Vinay(192425148)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary

*action. **Signature of the student:** Ch.B.Vinay*

Annexure A

Analytical Problem Solving

Q1. House No Area (sq.ft) Bedrooms Price (₹ Lakhs)

1 1000 2 50

2 1500 3 75

3 800 2 40

4 1200 3 60

5 2000 4 90

(i) Construct a linear regression model of the form

$$Price = \beta_0 + \beta_1(Area) + \beta_2(Bedrooms)$$

(ii) Estimate the regression coefficients.

(iii) Interpret the meaning of each coefficient.

Code:

```
from sklearn.linear_model import LinearRegression
import numpy as np
X = np.array([
    [1000, 2],
    [1500, 3],
    [800, 2],
    [1200, 3],
    [2000, 4]
])
y = np.array([50, 75, 40, 60, 90])
model = LinearRegression()
model.fit(X, y)
print(model.coef_)
print(model.intercept_)
print(model.predict([[1400, 3]]))
```

Output:

```
/Users/chitturibhaskarvinay/PyCharmMiscProject/.venv/bin/python /Users/chitturibhaskarvinay/PyC
[ 0.04252336 -0.28037383]
8.504672897196265
[67.19626168]

Process finished with exit code 0

mMiscProject > script12.py 15:
```

Q2.Email Word Frequency (Offer) Word Frequency (Win) Spam (1/0)

1 3 2 1

2 2 1 1

3 1 0 0

4 3 3 1

5 0 1 0

- (i) Construct a linear classification model.
- (ii) Determine the decision boundary.
- (iii) Classify a new email with: Offer = 2, Win = 1.

Code:

```
from sklearn.linear_model import LogisticRegression
import numpy as np
X=np.array([[3,2],[2,1],[1,0],[3,3],[0,1]])
y=np.array([[1],[1],[0],[1],[0]])
model = LogisticRegression()
model.fit(X,y)
new_email = [[2, 1]]
prediction = model.predict(new_email)
print("Prediction:", prediction[0])
if prediction[0] == 1:
    print("Spam Email")
else:
    print("Not Spam")
```

Output:

```
Prediction: 1
Spam Email

Process finished with exit code 0

mMiscProject > script12.py
```

Q3. Patient Fever (1/0) Headache (1/0) Disease (Yes/No)

1 1 1 Yes

2 1 0 Yes

3 0 1 No

4 1 1 Yes

5 0 0 No

(i) Compute prior probabilities.

(ii) Compute conditional probabilities.

(iii) Estimate the probability that a patient with Fever = 1 and Headache = 0 has the disease.

Code:

```
from sklearn.naive_bayes import BernoulliNB
import numpy as np
X = np.array([
    [1, 1],
    [1, 0],
    [0, 1],
    [1, 1],
    [0, 0]
])
y = np.array([1, 1, 0, 1, 0])
model = BernoulliNB()
model.fit(X, y)
patient = [[1, 0]]
prediction = model.predict(patient)
if prediction[0] == 1:
    print("Disease: Yes")
else:
    print("Disease: No")
prob = model.predict_proba(patient)
print("Probability:", prob)
```

Output:

```
/Users/chitturibhaskarvinay/PyCharmMiscProject/.venv/bin/python /Users/chitt  
Disease: Yes  
Probability: [[0.20661157 0.79338843]]  
  
Process finished with exit code 0
```